

Unit 5, Lesson 2: Scale Factors – Perimeter & Area

Benchmark

MA.7.GR.1.5 Solve mathematical and real-world problems involving dimensions and areas of geometric figures, including scale drawings and scale factors.

Learning Targets

- ☐ I can relate the perimeters and areas of rectangles to their scale factors.
- ☐ I can relate the perimeters and areas of triangles to their scale factors.
- ☐ I can relate the perimeters and areas of trapezoids to their scale factors.

5.2.1 Warm-Up: World's Largest Gummy Bear

The World's Largest Gummy Bear is the equivalent of 1,400 regular sized gummy bears and packs a whopping 6,120 calories.

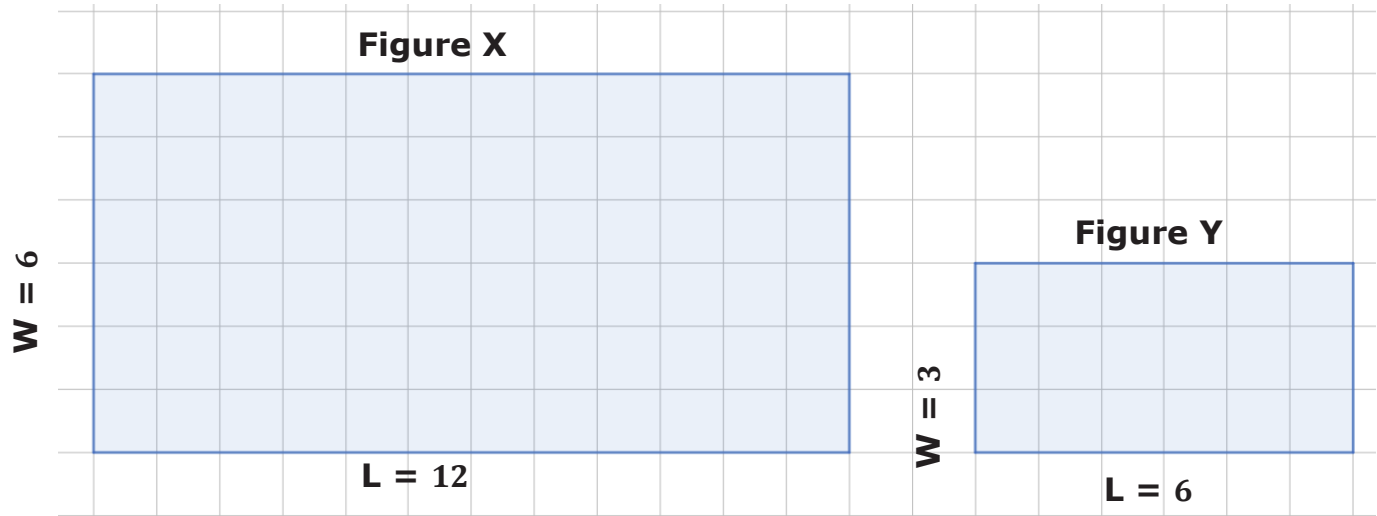


1. What do you wonder about the World's Largest Gummy Bear?
2. The height of the World's Largest Gummy Bear is $9\frac{1}{2}$ inches (in.). Guess what the scale factor for the World's Largest Gummy Bear is to a normal gummy bear.
3. The height of a regular gummy bear is 0.8 in. Use your scale factor to determine the accuracy of your guess.

5.2.2 Exploration: Scale Factor and Perimeter

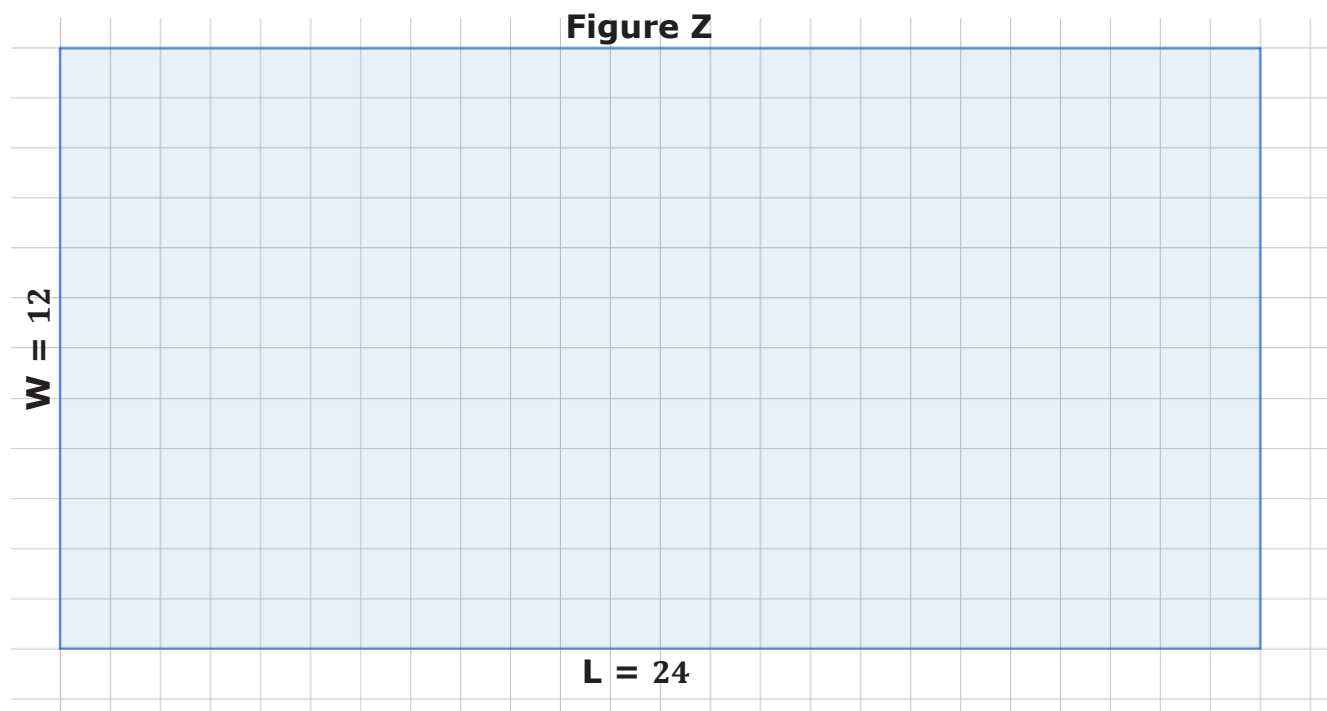
Work with your partner to complete the following.

1. Paolo drew Figure X to represent a picture he's putting in an advertisement he is designing. Paolo's rectangle has a length (L) of 12 units by a width (W) of 6 units. Because he has several design options for his advertisement, he also drew a scale model, Figure Y, with a length (L) of 6 units and a width (W) of 3 units.



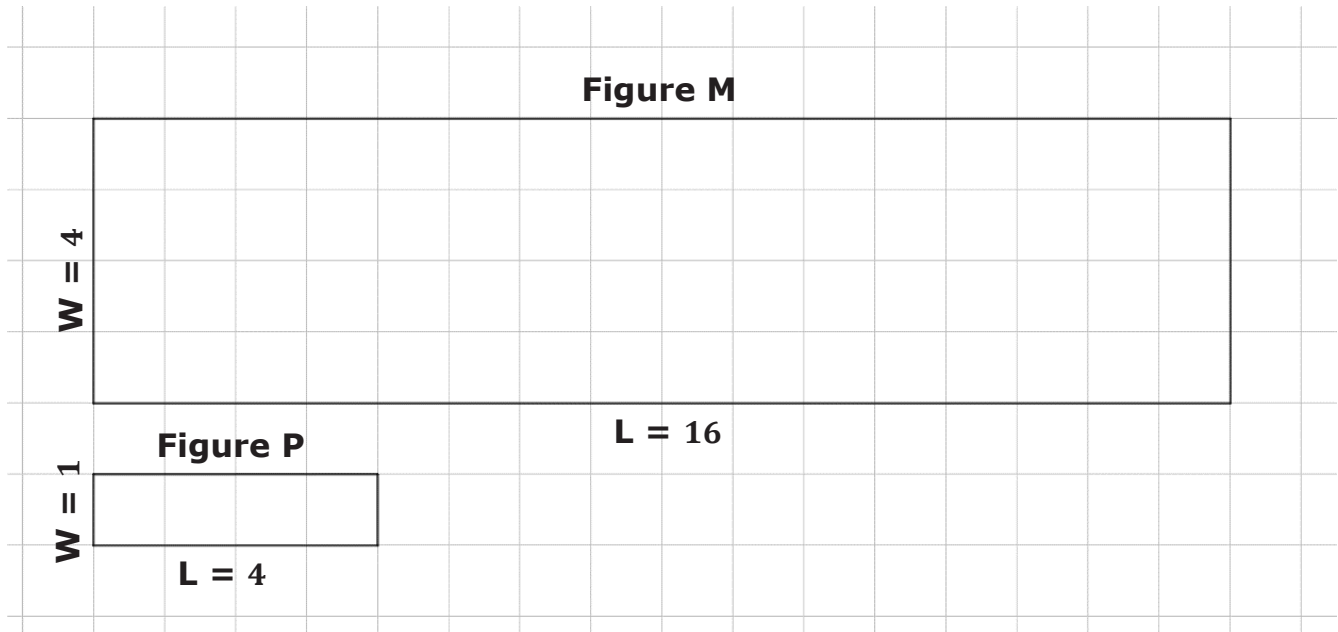
- a. The scale factor from Figure X to Figure Y is $\frac{1}{2}$.
- b. What is the perimeter of Figure X?
- c. What is the perimeter of Figure Y?
- d. What is the ratio of the perimeter of Figure Y to the perimeter of Figure X?

Paolo decided to change the scale factor to 2 to create Figure Z from Figure X. Figure Z has a length (L) of 24 units by a width (W) of 12 units.



- e. What is the perimeter of Figure Z?
- f. How does that compare to the perimeter of Figure X?
- g. What relationship, if any, is there between the scale factor and the ratio of the perimeters?

2. Figure M has a length (L) of 16 units and a width (W) of 4 units. Figure P is a scale model of Figure M. Figure P has a length (L) of 4 units and a width (W) of 1 unit.

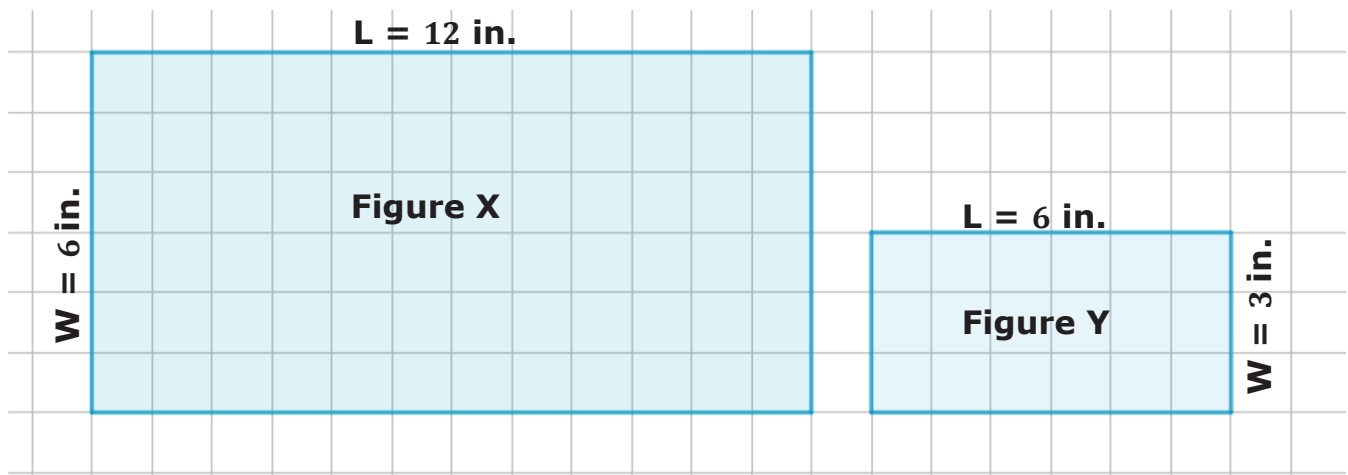


- a. The scale factor from Figure M to Figure P is $\frac{1}{4}$.
- b. What is the perimeter of Figure M?
- c. Without finding the sum of the sides, what do you expect the perimeter of Figure P to be?
- d. Verify your answer by finding the sum of the sides of Figure P.
3. With your partner, write a rule that can be used to calculate the perimeter of a scale model if given only an original perimeter and a scale factor.

5.2.3 Exploration: Scale Factor and Area

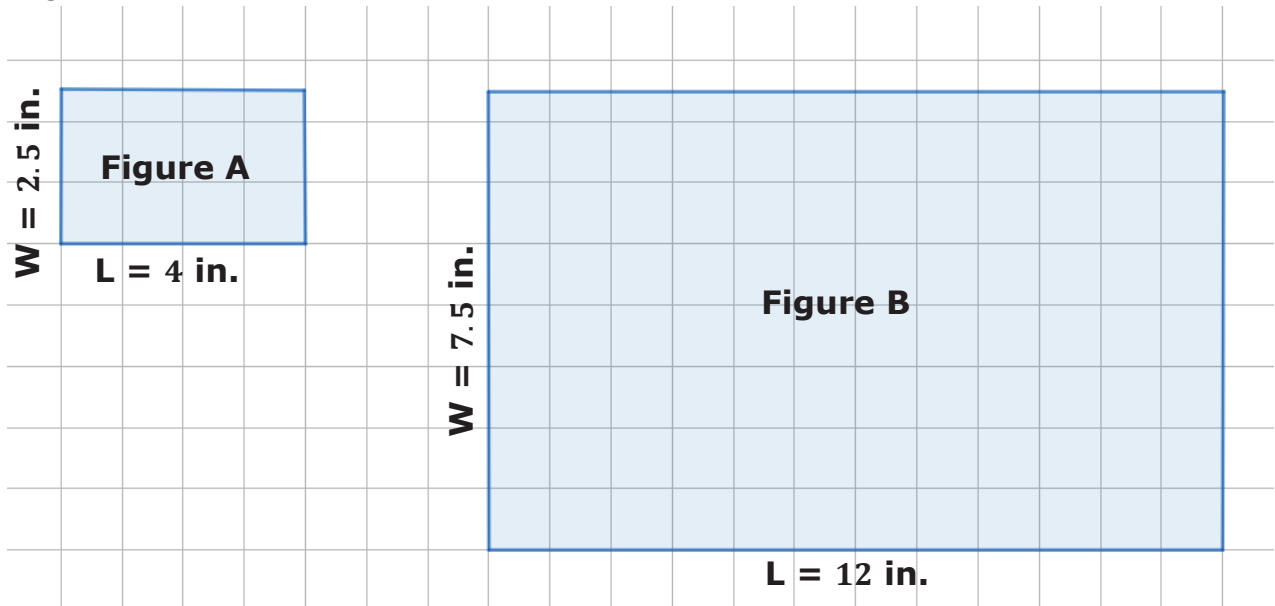
Work with your partner to complete the following.

1. The same figures from Paolo's advertisement are shown where Figure X has a length (L) of 12 units and a width (W) of 6 units, and Figure Y has a length (L) of 6 units and a width (W) of 3 units.



- a. The scale factor from Figure X to Figure Y is _____.
- b. What is the area of Figure X?
- c. What is the area of Figure Y?
- d. What is the ratio of the area of Figure Y to the area of Figure X?
- e. What relationship, if any, is there between the scale factor and the ratio of the areas?

2. Figure B is a scale model of Figure A. Figure B has a length (L) of 12 inches (in.) and a width (W) of 7.5 in. Figure A has a length (L) of 4 in. and a width (W) of 2.5 in.

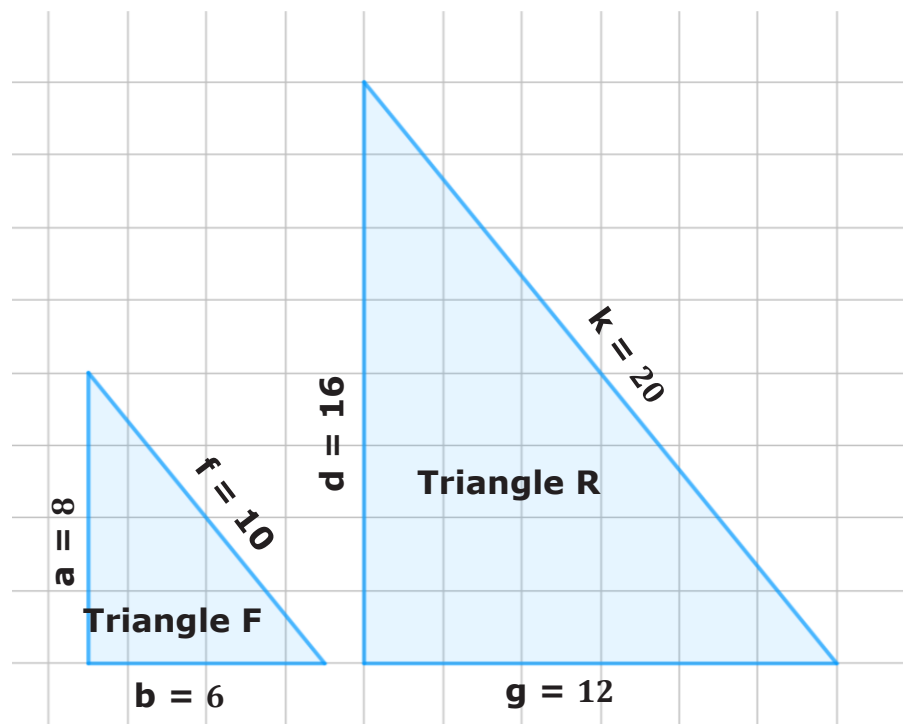


- What is the scale factor from Figure A to Figure B?
- What is the area of Figure A?
- What is the area of Figure B?

- d. How does the scale factor from Figure A to Figure B relate to the ratio of the area of Figure A to Figure B?
 - e. Discuss the relationship between the scale factor and the area you observed in problems 1 and 2.
3. With your partner, write a rule that can be used to calculate the area of a scale model if given only an original area and a scale factor.

5.2.4 Guided Instruction: Comparing Perimeter and Area of Scaled Triangles and Trapezoids

1. Right Triangle F has a base (b) of 6 units, a height (a) of 8 units, and a third side (f) of 10 units. Right Triangle R is a scaled image of Triangle F. Triangle R has a base (g) of 12 units, a height (d) of 16 units, and a third side (k) of 20 units.

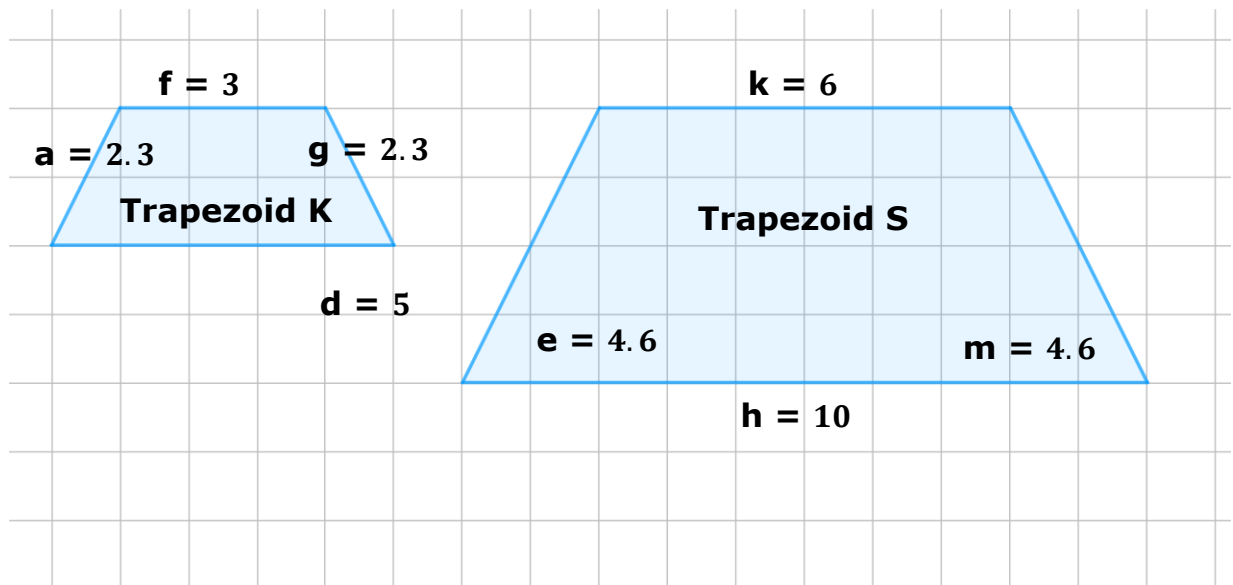


- a. The scale factor from Triangle F to Triangle R is _____.
- b. Compare the perimeter of Triangle F to the perimeter of Triangle R.
- c. Using Triangle F and the listed scale factors, complete the table.

Scale Factor	Base (cm)	Height (cm)	3 rd side (cm)	Perimeter (cm)	Area (cm ²)
1	6	8	10	24	24
3	18	24			
$\frac{1}{2}$	3		5		

- d. Discuss with your partner whether the rules you wrote for calculating the perimeter or area of a scaled rectangle, given a scale factor and the original perimeter or area, apply to triangles.

2. Trapezoid K has two bases. One base (d) is 5 units long and the other 3 (f) units long. The other two sides (a and g) have equal lengths of 2.3 units. A scale factor of 2 has been applied to Trapezoid K resulting in Trapezoid S. Trapezoid S has two bases, one (h) 10 units long and the other (k) 6 units long. The other two sides (e and m) have equal lengths of 4.6 units.



- a. How does the perimeter of Trapezoid K compare to the perimeter of Trapezoid S?
- b. Explain how these perimeters relate to the scale factor.
- c. How should the areas of the figures relate?

5.2.5 Wrap-Up: Cool Down

1. Write a note to a classmate who missed today's lesson. Include academic vocabulary related to scale factors.

